TPRM Platform: one platform, one database, one model gateway, and the agents inside the AI worker

Zoom out for the platform. Zoom into the AI worker box for the orchestrator, the agents, the three jobs, and the token controls. Screens and the API never call a model; only the worker does, through Tachyon. Agent names mirror the ProcessUnity vocabulary so the two can be compared side by side.

Internal users
PBL, SMC, RM, Admin
Browser, bank SSO

Suppliers
Portal, separate identity
Upload, answer, confirm

React web app
tprm/web

Nine screens, navigation by role
PDF viewer with highlights
Talks only to the API

FastAPI API
tprm/api

SSO, roles, tenant context
Records, approvals, decisions
Usage API for the Admin screen
Enqueues jobs, never calls a model

DATA

Postgres + pgvector
Records, chunks, vectors, full-text
Job queue and immutable run log
Row-level security per tenant

Object storage
Original documents by SHA-256
Shared across lines of business
Encrypted at rest

AI worker · tprm/ai_worker

Everything the worker does. Sequence only where a step needs the previous result; everything else fans out in parallel, up to the tenant's cap, after one warm-up call that fills the prompt cache.

1 · HOW AN AGENT IS CREATED

Agent template
one definition file per agent in the prompt library
Name, and the one job it does · Prompt version · Output schema, JSON, checked in code · Tools it may call · Model tier: small, mid, large on escalation · Budget and rate limit · Confidence threshold · Kill switch
New agent = one file plus a passed evaluation set. No code change.

AGENT REGISTRY · TWENTY AGENTS FROM THE TEMPLATE · NAMES MIRROR PROCESSUNITY'S

Duplicate Third Party Inspector small · ranks unclear matches	**Duplicate Service Inspector** small · duplicate service check	**Entity Verifier** code · GLEIF legal entity check	**Risk Signal Summarizer** small · 90-day news, cached
IRQ Responder mid · suggested IRQ answers	**Trust Center Finder** small · public certificates	**Document Intake Checker** small · type, entity, period	**Document Processor** code · extract, chunk, embed
SOC 2 Analyzer mid · scope, exceptions, CUECs	**Evidence Mapper** mid · controls to passages	**Escalation Reviewer** large · uncertain items only	**Findings Drafter** mid · one draft per gap
Issue Remediator mid · drafts supplier request	**Contract Scanner** mid · key terms to clauses	**Insurance Guardian** small · limits vs thresholds	**Feed Collector** code · nightly pulls to cache
Alert Triage small · dedupe and rank	**Digest Writer** small · nightly, all sourced	**Executive Risk Reporter** small · one-page summaries	**Ask Your Data** mid · query, validated in code

2 · HOW AGENTS RUN FOR THREE REAL JOBS · WHITE = AGENT (ONE MODEL TIER) · DASHED = CODE, ZERO TOKENS · STACKED = RUNS IN PARALLEL · GREEN = A PERSON

Orchestrator
one per job

Creates each agent from its template

Hands it the job and only the data it needs

Runs the fixed graph: waits where a step depends on another, fans out where it does not

One warm-up call, then the fan-out, up to the tenant's cap

Stops the chain when a check fails

Pauses for the person, then resumes

Writes one run record per call

Job A · A supplier uploads a SOC 2 report · overnight batch · about $0.45 in the example

Document Intake Checker — small tier
Hash, type, entity, period. Seen before? Reuse, stop.

Document Processor — code, 0 tokens
Extract, chunk, embed on CPU. Stored by hash.

×10 in parallel
Evidence Mapper — mid tier, batch
Warm-up call, then 10 controls per call on the cached text.

Verifier — code, 0 tokens
Schema check. Quote must exist in its chunk. Flags low confidence.

×8 in parallel
Escalation Reviewer — large tier
Only the uncertain controls, 2K tokens each.

×10 in parallel
Findings Drafter — mid tier
One draft per gap: quote, severity, fix.

Person approves — SMC
Accept, edit, reject. Only then a record changes. Logged.

Job B · An SMC types a third-party name · runs while the user waits · under 10K tokens

Database search — code, 0 tokens
Names, aliases, LEI, domain. Exact and fuzzy.

all three at the same time, then joined

Duplicate Third Party Inspector — small tier
Ranks only the unclear matches, with a reason.

Entity Verifier — code, 0 tokens
GLEIF: legal name, LEI, parent.

Risk Signal Summarizer — small, daily cache
90-day feeds, each line sourced.

Verifier — code, 0 tokens
A line with no source link is dropped.

Person picks — SMC
Uses a record or creates one. Logged.

Job C · Every night, nobody waiting · one batch per portfolio · small tier only

×4 feeds in parallel
Feed Collector — code, 0 tokens
Pulls news, ratings, finance, sanctions to cache.

×portfolios in parallel
Alert Triage — small tier
Dedupes 61 alerts to 32, ranks by the service we buy.

×portfolios in parallel
Digest Writer — small tier
Daily digest; every item links to its source.

Executive Risk Reporter — small tier
One-page summaries, cached until data changes.

Person reviews — SMC, RM, morning
Acts on 3, reads 7, the rest is logged.

3 · WHAT EVERY AGENT SHARES, AND WHAT IT COSTS

Shared tools
Retrieval over the chunk store, on CPU
Registry lookups (GLEIF), feed cache
Tachyon router: tiers, prompt cache, batch
Run log writer, one record per call
No agent calls another on its own

Token controls
Analyse once per document hash, reuse everywhere
Text, not images; keep only relevant sections
Cache the stable prefix; batch overnight at half price
Small or mid model first; escalate the uncertain few
JSON schema, capped output; top passages for questions
Warm the cache once, then fan out to the cap

Guardrails
One job per agent; the orchestrator fixes the order
Document text is data, never an instruction
Output checked in code before anyone sees it
Budget, rate limit, kill switch per agent and tenant
Prompt version and model logged on every call

People decide
Agents draft; a named person accepts, edits, rejects
Nothing changes a record or reaches a supplier without a click
Acceptance rate is the accuracy number we watch
A low rate sends the agent back to its evaluation set

THE NAIVE WAY
Send the whole document with every question: 100 × 120K = 12M tokens.
About $24 per assessment, per tenant

THIS WORKER
Cached, batched, verified, escalated only when unsure, fanned out up to the cap.
About $0.45 for the same work

SAME SUPPLIER, NEXT TENANT
Another line of business onboards the same supplier with the same document.
$0 served from the store

Tachyon AI gateway
Claude, GPT, Gemini
Prompt cache, batch
The only model endpoint

nightly

Feeds and registries
News, cyber ratings, financial health, sanctions, GLEIF
Pulled nightly, cached

Estimates at public list prices; Tachyon's internal rates may differ. Two-headed arrows are request and response; the dashed arrow is a nightly pull; the highlighted arrow is the only path on which a model is called. Agent names follow ProcessUnity's vocabulary (IRQ, third party, service, issue, remediation) for side-by-side comparison; the platform itself is ours.

PLATFORM

OpenShift
Three deployables plus a CPU model pod; configuration per environment

Observability
OpenTelemetry traces and metrics; the ai_run table is the source of truth for the Admin screen

CI/CD and evaluation gate
Lint, types, tests, DeepEval; a prompt or model change below its gate cannot ship

Identity and secrets
OIDC single sign-on, roles from groups, secrets injected from the vault